HS-401: Professional Values and Ethics

LECTURE NO 8: *Designing Morality*

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Contents

1.1 Introduction (1/3)

Case Robert Moses' Racist Overpasses

1.1 Introduction (2/3)

Robert Moses (1888–1981) was a very influential and also contested urban planner of mid-twentieth century New York. In his 1974 book, *The Power Broker*, biographer Robert Caro argued that Moses also demonstrated racist tendencies. He designed several overpasses over the parkways on Long Island, which were too low to accommodate buses. Only cars could pass below them and for that reason the overpasses complicated access to Jones Beach Island. Only people who could afford a car – and in Moses' days these were generally not Afro-American people – could easily access the beaches now.

This case has become especially famous due to the philosopher of technology Langdon Winner who mentioned it in his article "Do artifacts have politics?" (1980). As Winner notes although the overpasses are extraordinary low, one would normally not be inclined to attach any special meaning to that fact. It turns out, however, that they were deliberately designed to achieve a specific social effect. In the words of Langdon Winner:

> that would discourage the presence of buses on his parkways. According to … Moses' biographer, Robert A. Caro, the reasons reflect Moses' social class bias and racial prejudice. Automobile-owning whites of "upper" and "comfortable middle" classes … would be free to use the parkways for recreation and commuting. Poor people and blacks, who normally used public transit, were kept off the roads because the twelve-foot tall buses could not handle the overpasses.

1.1 Introduction (3/3)

- It was actually possible to reach the beach by bus, bypassing the overpasses. The example shows that technological artifacts can be politically or morally charged. This has implications for the ethics of design because it charges engineers with a responsibility to think about the potential moral and political role of artifacts in the design process.

1.2 Ethics as a Matter of Things (1/2)

- As the example of the "racist" overpasses shows, the artifacts we deal with in our daily lives help to determine our actions and decisions in myriad (so many) ways.
- We should not consider morality as a solely human affair, but also as a matter of *things*.
- Langdon Winner's analysis of Moses' Long Island overpasses dates from 1980. In this case, allegedly one man was able to embed his racial ideology within these technological artifacts, thereby racializing their construction and eventual use.
- The ethics of engineering design is the best place to start analysing the moral dimension of technological artifacts, since this is also the place where human beings can take responsibility for the moral aspects of their products.
- It mainly focuses on the importance of taking individual responsibility (e.g., "whistle blowing") to prevent technological disasters, and on methods to assess and balance the risks accompanying new technologies.

1.2 Ethics as a Matter of Things (2/2)

- For analysing these impacts of technologies, and the various aspects of the role technological artifacts play in their use contexts, the concept of *technological mediation* is a helpful tool.
- *Technological mediation* refers to the ways in which technology acts as an intermediary or mediator in various aspects of human life, communication, and interaction. This concept encompasses the idea that technology plays a central role in shaping, influencing, and facilitating our experiences with the world.

1.3 Technological Mediation (1/2)

- Technological artifacts help to shape human ***actions*** and ***perceptions***, and create new practices and ways of living. Cell phones, for example, contribute explicitly to the nature of our communications and interactions. Technologies like ultrasound play active roles in our decisions regarding unborn life.
- When technologies fulfil their functions, they also help to shape the actions and experiences of their users. This phenomenon is called ***technological mediation***.

1.3.1 Mediation of perception

- "*The influence of artifacts on human perception, that is, the sensory relationship with reality*."
- A thermometer, for instance, establishes a relationship between humans and reality in terms of temperature. Reading off a thermometer does not result in a direct sensation of heat or cold, but gives a value which requires interpretation in order to tell something about reality.

1.3 Technological Mediation (2/2)

1.3.2 Mediation of action

- "*The influence of artifacts on human action*."
- A prescription how to act that is built (designed) into an artifact is called as a *script*.
- The influence of artifacts on human actions has a specific nature.
- A traffic sign (signal) makes people slow down because of what it signifies, not because of its material presence in the relation between humans and world.

1.4 Moralizing Technology (1/3)

- As discussed earlier, many of our actions and interpretations of the world are co-shaped by the technologies we use. *Telephones* mediate the way we communicate with others, *cars* help to determine the acceptable distance from home to work, *thermometers* co-shape our experience of health and disease.
- The mediating role of technologies also pertains to actions and decisions we usually call "moral". If ethics is about the question "how to act," and technologies help to answer this question, technologies appear to do ethics, or at least help us to do so.
- Technologies do not only enable us to perform actions and have experiences that were rarely possible before, but in doing so, they also help to shape how we act and experience things. Artifacts help to shape human actions, interpretations, and decisions, which would have been different without the artifact.
- Quite often, technologies mediate human actions and experiences without human beings having told them to do so. Some technologies, for instance, are used differently than their designers had imagined. The first cars – which only made 15 km/h – were used primarily for sports, and for medical purposes.

1.4 Moralizing Technology (2/3)

1.4.1 Criticizing the moral character of technological artifacts

- The idea that artifacts have morality has been severely criticized. A main criticism is that mediation has nothing to do with morality whatsoever. Not only are technological artifacts unable to make moral decisions, but also human-induced technology behaviour not has a moral character.
- Should all human actions be guided by technology, the criticism goes, the outcome would be a technocratic society in which moral problems are solved by machines instead of people.
- Actions that are not the product of our own free-will but are induced by technology cannot be described as "moral."

Examples of Moralizing Artifacts

Speed bump: "Lower your driving speed"
Metro tourniquets: "Pay for public transport"
Hotel keys (with large object): "Return your hotel key to the desk"
Door-closer: "Close the door"
Alcohol lock for car (car lock that analyses your breath): "Don't drive drunk"

1.4 Moralizing Technology (3/3)

1.4.2 Taking mediation into ethics

- ***Moralization of technology-*** The deliberate development of technologies in order to shape moral action and decision-making.

1.5 Designing Mediations (1/4)

- The moralization of technological artifacts is not as easy as it might seem to be. In order to "build in" specific forms of mediation in technologies, designers need to anticipate the future mediating role of the technologies they are designing. And this is a complex task, since there is no direct relationship between the activities of designers and the mediating role of the technologies they are designing.
- Technologies, however, can be used in unforeseen ways, and therefore have an unforeseen influence on human actions. The classical energy-saving light bulb is a good example here, having actually resulted in an increased energy consumption since such bulbs often appear to be used in places previously left unlit, such as in the garden or on the façade, thereby cancelling out their economizing effect.

1.5 Designing Mediations (2/4)

- Designers thus help to shape the mediating roles of technologies, but these roles also depend on the ways in which the technologies are used .
- Designers cannot simply "inscribe" a desired form of morality into an artifact. The mediating role of technologies is not only the result of the activities of the designers, who inscribe scripts or delegate responsibilities, but also depends on the users, who interpret and appropriate technologies.
- Figure 1 next slide illustrates these complicated relations between technologies, designers, and users in the mediation of actions and interpretations.

1.5 Designing Mediations (3/4)

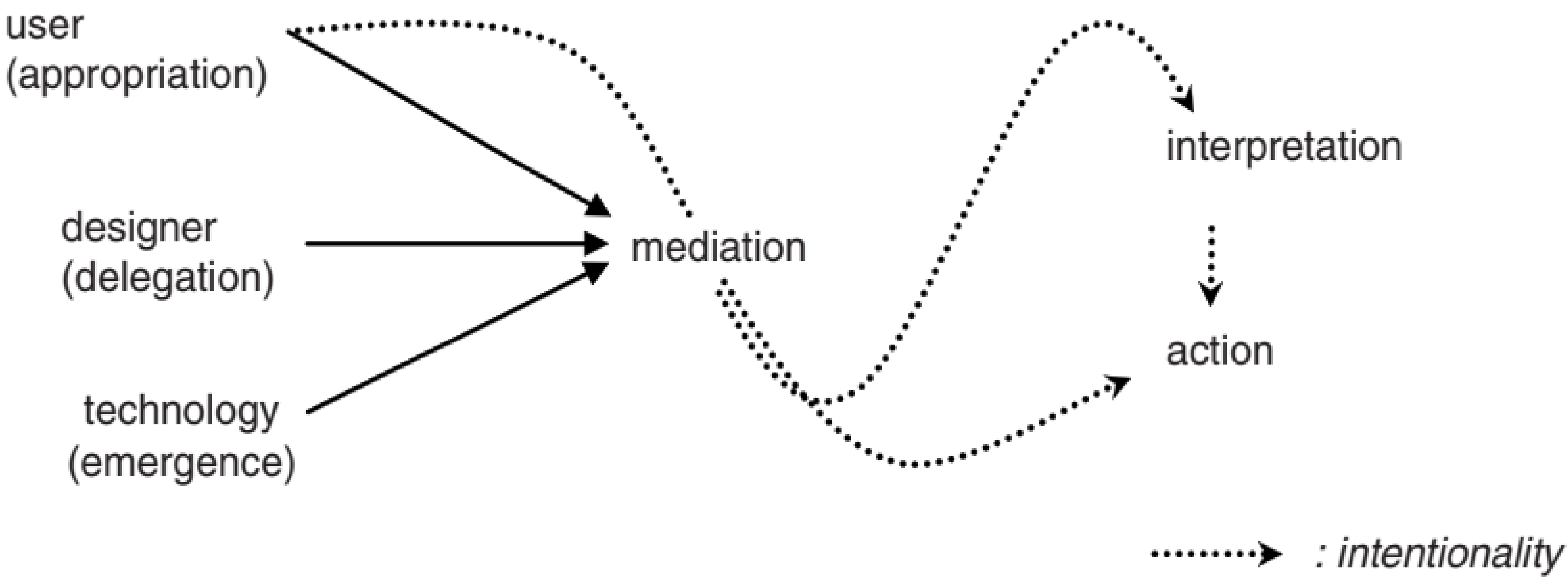

Figure 1: Human actions and interpretations informing moral decisions

1.5 Designing Mediations (4/4)

- It is important to seek links between the design context and the future use context. Design specifications should be derived not only from the product’s intended function but also from an informed prediction of the product’s mediating roles and a moral assessment of these roles. A key tool to bringing about this coupling of design context and use context, however trivial it may sound, is the designer’s moral imagination.
- ***Anticipating mediation by imagination-***Trying to imagine the ways technology-in-design could be used. This insight is then used to deliberately shape user operations and interpretations.

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